

PROJECT SPECIFICATION DOCUMENT FOR MSCI 546 - ADVANCED MACHINE LEARNING

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Project (P): Team-based Course Project (30%). Each student will work on a machine learning project as part of a team. Teams will work on Kaggle competitions or similar. Teams are required to implement at least one neural network-based model as part of their exploration. Grading will be based on a) model performance, b) insights drawn from data (via exploratory data analysis). Refer to course outline for the deadlines associated with this component.

P1. Team Formation

- Students will form 5 person teams and communicate to the teaching staff by **end of week 1** on Course Campuswire.
- All students who are not a part of a team by this date will be randomly assigned to teams.
- Final teams assignments along with a Team Number (randomly assigned) will be announced on Campuswire; It is team/student's responsibility to get back to us with any discrepancy regarding team memberships within two days of this announcement.
- The teaching staff reserves the right to adjust team compositions to ensure the best learning experience for all students. This flexibility allows us to create a balanced and productive environment for everyone involved, especially after the end of Add/Drop period.

P2. Proposal and Baseline: For the proposal stage Teams will compile a 2 page pdf document outlining the following. The second page can only contain any figures or references.

- Identify a Kaggle competition or a project they wish to participate in along with the link; Teams can participate in close competitions as well.
- Identify the type of machine learning (ML) task (classification, regression, etc.) this will include;
- Analyze the feasibility of the project (can these be trained locally or on resources available at the university, is the data amenable for developing neural network-based models?);
- Share any insights from their exploratory data analysis (EDA);
- Identify metrics they will use to analyze performance;
- Identify and implement a **simple baseline**, and report the corresponding performance metrics.

Exploratory data analysis (EDA) is an investigative approach to summarizing data sets' main characteristics using visualizations and statistics, uncovering patterns, outliers, and relationships before formal modeling.

A **baseline** in ML is a simple, basic model or method used as a reference point to evaluate other models' performance.

P3. Development Stage: Teams will implement different ML models and feature engineering to improve performance over the simple baseline. Teams are expected to implement/try at least one method a week based on the learning from the previous week. Additional guidelines are as follows:

- This is the main stage of the project, at the end of this stage Teams must have at least 1 method per team member (eg. 5 in the case of a 5 member team including the initial baseline submitted at the proposal stage) plus a neural network-based method (all members can work on this) – so 6 in total. Based on these methods the team will propose their solution for the task.
- Each member is required to implement one method.
- There are no deliverables for this phase.

P4. Slides-style Report and P5. In-class presentations: Based on the models teams implement, they will submit a slides-style report as a pdf (20 slides max. including references). These slides (or a part of these slides) should be used for the final in-class presentation. No additional materials can be added to the slides for the final presentation. The slides (report and presentation) can cover interesting aspects of the project, and learning, and must include

- Task
- Data
- Metrics
- Baselines (at least four including the initial baseline)
- Their proposed solution
- Insights into the properties of their solutions (visualizations, interpretations)
- Improvement over baselines

Quality of slides, visualizations, plots, types of models used, improvement over baselines, and key takeaways will be used for evaluation. Difficulty of task will be considered when evaluating improvement over baseline(s).

Presentation Length The presentation should be at most 8 minutes long, plus 1-2 minutes Q&A. The 8 minute duration is a hard stop to keep the presentations within the class time. *Note that a team may be asked to present in the next lecture slot if for any reason we run over class time.* This means that you are expected to be ready to present in the next class if required.